The Cactus Midpoint Theorem

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Abstract

We prove that in the 0-Hecke algebra $H_0(S_k)$, the Tikhonov interpolating operators $R_i(p) = pT_i + (1-p)$ evaluated at $p = 1/2$ yield a scalar multiple of the cactus (Demazure) sorting operator: $\prod R_{i_j}(1/2) = (1/2)^m \pi_{w_0^{[a,b]}}$. We show this identity is *special to* $q = 0$ by proving that the Demazure elements $\pi_i = T_i + 1$ in $H_q(S_k)$ satisfy a braid obstruction $\pi_i\pi_{i+1}\pi_i - \pi_{i+1}\pi_i\pi_{i+1} = q(T_i - T_{i+1})$, which vanishes precisely at $q = 0$. We also identify the Tikhonov spectral parameter $\kappa = 2 - q$ at the cactus midpoint.

1 Setup

Let $H_q(S_k)$ be the Iwahori–Hecke algebra of type A_{k-1} over $\mathbb{Q}(q)$, with generators $T_1, \ldots, T_{k-1}$ satisfying:

$$T_i^2 = (q-1)T_i + q, \tag{1}$$

$$T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}, \tag{2}$$

$$T_iT_j = T_jT_i \quad \text{for } |i-j| \geq 2. \tag{3}$$

Definition 1. The *Demazure element* (or π-operator) is $\pi_i := T_i + 1$.

Definition 2. The *Tikhonov interpolating operator* is $R_i(p) := pT_i + (1-p) \in H_q(S_k)[p]$.

For an interval $[a,b] \subseteq [1,k]$, let $w_0^{[a,b]}$ denote the longest element of the parabolic subgroup $S_{[a,b]}$, with $m = \ell(w_0^{[a,b]}) = \binom{b-a+1}{2}$.

2 The Demazure Algebra

Lemma 3 (Quasi-idempotency)**.** *In $H_q(S_k)$, the Demazure elements satisfy*

$$\pi_i^2 = (q+1)\,\pi_i.$$

In particular, π_i is idempotent if and only if $q = 0$.

Proof. Expanding:

$$\begin{aligned}\pi_i^2 = (T_i+1)^2 &= T_i^2 + 2T_i + 1 \\ &= \big((q-1)T_i + q\big) + 2T_i + 1 \\ &= (q+1)T_i + (q+1) \\ &= (q+1)(T_i+1) = (q+1)\,\pi_i.\end{aligned}$$

At $q = 0$ this gives $\pi_i^2 = \pi_i$. □

Theorem 4 (Braid Obstruction)**.** *For adjacent generators π_i, π_{i+1} in $H_q(S_k)$:*

$$\pi_i\pi_{i+1}\pi_i - \pi_{i+1}\pi_i\pi_{i+1} = q(T_i - T_{i+1}). \tag{4}$$

The braid relation for π_i holds if and only if $q = 0$.

Proof. We expand both sides using the Hecke relations.

Left side. First compute $(T_{i+1} + 1)(T_i + 1) = T_{i+1}T_i + T_{i+1} + T_i + 1$. Then:

$$\begin{aligned}
\pi_i\pi_{i+1}\pi_i &= (T_i + 1)\big(T_{i+1}T_i + T_{i+1} + T_i + 1\big) \\
&= T_iT_{i+1}T_i + T_iT_{i+1} + T_i^2 + T_i + T_{i+1}T_i + T_{i+1} + T_i + 1 \\
&= T_iT_{i+1}T_i + T_iT_{i+1} + T_{i+1}T_i + T_{i+1} + (q+1)T_i + (q+1),
\end{aligned}$$

where we used $T_i^2 = (q-1)T_i + q$ to get the last line: $T_i^2 + T_i = (q-1)T_i + q + T_i = qT_i + q$, and the remaining T_i gives $(q+1)T_i + (q+1)$ total.

Right side. By symmetry under $i \leftrightarrow i+1$:

$$\pi_{i+1}\pi_i\pi_{i+1} = T_{i+1}T_iT_{i+1} + T_{i+1}T_i + T_iT_{i+1} + T_i + (q+1)T_{i+1} + (q+1).$$

Difference. Using the braid relation $T_iT_{i+1}T_i = T_{i+1}T_iT_{i+1}$, the leading terms cancel:

$$\begin{aligned}
\pi_i\pi_{i+1}\pi_i - \pi_{i+1}\pi_i\pi_{i+1} &= \big[T_{i+1} + (q+1)T_i\big] - \big[T_i + (q+1)T_{i+1}\big] \\
&= qT_i - qT_{i+1} \\
&= q(T_i - T_{i+1}).
\end{aligned}$$

The terms $T_iT_{i+1} + T_{i+1}T_i$ appear identically on both sides. □

Corollary 5 (Word-Independence at $q = 0$)**.** *At $q = 0$, for any $w \in S_k$, the product $\pi_w := \prod_{j=1}^m \pi_{i_j}$ over a reduced word $s_{i_1} \cdots s_{i_m}$ of w is independent of the choice of reduced word.*

Proof. The reduced words for w are connected by braid moves ($s_is_{i+1}s_i \leftrightarrow s_{i+1}s_is_{i+1}$) and commutation moves ($s_is_j \leftrightarrow s_js_i$ for $|i-j| \geq 2$) by Matsumoto's theorem. By Theorem **??**, the braid relation $\pi_i\pi_{i+1}\pi_i = \pi_{i+1}\pi_i\pi_{i+1}$ holds at $q = 0$. The commutation relation $\pi_i\pi_j = \pi_j\pi_i$ for $|i-j| \geq 2$ follows from (**??**). Hence the product is word-independent. □

Corollary 6 (Word-Dependence at Generic q)**.** *At generic $q \neq 0$, the product $\prod \pi_{i_j}$ over a reduced word for w_0 depends on the choice of reduced word. In the regular representation of $H_q(S_3)$, the product $(T_1 + 1)(T_2 + 1)(T_1 + 1)$ has rank 3, not rank 1. It is not a cactus commutor.*

Proof. The obstruction $q(T_1 - T_2) \neq 0$ for $q \neq 0$ shows word-dependence. For the rank claim, the product acts on the 6-dimensional regular representation. At $q = 0$, idempotency forces the image of each π_i to be a proper subspace, and the composition collapses to rank 1. At $q = 1$ (the symmetric group), direct computation shows rank 3 (verified computationally in $H_1(S_3)$). Since the rank is an upper semicontinuous function of q, the rank is ≥ 3 at generic q. □

3 The Cactus Midpoint Theorem

Lemma 7 (Factorization of $R_i(p)$)**.** *The Tikhonov operator factors as:*

$$R_i(p) = p\,\pi_i + (1-2p)\cdot \mathrm{id}\,. \tag{5}$$

Proof. $R_i(p) = pT_i + (1-p) = p(T_i+1) - p + (1-p) = p\,\pi_i + (1-2p)$. □

Theorem 8 (Cactus Midpoint)**.** *At $q=0$, for any interval $[a,b] \subseteq [1,k]$ and any reduced word $s_{i_1}\cdots s_{i_m}$ of $w_0^{[a,b]}$:*

$$\prod_{j=1}^{m} R_{i_j}\big(\tfrac{1}{2}\big) = \big(\tfrac{1}{2}\big)^m \pi_{w_0^{[a,b]}}. \tag{6}$$

Proof. By Lemma **??**, at $p=1/2$:

$$R_i\big(\tfrac{1}{2}\big) = \tfrac{1}{2}\,\pi_i + 0 = \tfrac{1}{2}\,\pi_i.$$

The identity term $(1-2p)$ vanishes at $p=1/2$. Therefore:

$$\prod_{j=1}^{m} R_{i_j}\big(\tfrac{1}{2}\big) = \prod_{j=1}^{m} \tfrac{1}{2}\,\pi_{i_j} = \big(\tfrac{1}{2}\big)^m \prod_{j=1}^{m} \pi_{i_j} = \big(\tfrac{1}{2}\big)^m \pi_{w_0^{[a,b]}},$$

where the last equality uses word-independence (Corollary **??**) and the definition $\pi_{w_0^{[a,b]}} = \prod \pi_{i_j}$. □

Theorem 9 (Möbius Factorization)**.** *For the product over a reduced word of $w_0^{[a,b]}$ at $q=0$:*

$$\prod_{j=1}^{m} R_{i_j}(p) = \sum_{S\subseteq\{1,\ldots,m\}} p^{|S|}(1-2p)^{m-|S|}\cdot \vec{\pi}_S, \tag{7}$$

*where $\vec{\pi}_S = \prod_{j\in S}^{\rightarrow} \pi_{i_j}$ is the ordered product of π_{i_j} for $j\in S$ (in the order inherited from the reduced word), and identity operators fill positions $j\notin S$. At $p=1/2$, only the term $S=\{1,\ldots,m\}$ survives, recovering Theorem **??**.*

Proof. By Lemma **??**, $R_{i_j}(p) = p\,\pi_{i_j} + (1-2p)\cdot\mathrm{id}$. Expanding the product:

$$\prod_{j=1}^{m}\big(p\,\pi_{i_j} + (1-2p)\cdot\mathrm{id}\big) = \sum_{S\subseteq\{1,\ldots,m\}} \prod_{j=1}^{m} \begin{cases} p\,\pi_{i_j} & j\in S \\ (1-2p)\cdot\mathrm{id} & j\notin S \end{cases} = \sum_{S} p^{|S|}(1-2p)^{m-|S|}\,\vec{\pi}_S.$$

At $p=1/2$, $(1-2p)=0$, so only $S=\{1,\ldots,m\}$ contributes: $p^m\vec{\pi}_{\{1,\ldots,m\}} = (1/2)^m\,\pi_{w_0^{[a,b]}}$. □

4 The Eigenvalue Picture

Proposition 10 (Eigenvalues of $R_i(p)$)**.** *The generator T_i has eigenvalues q (symmetric eigenspace) and -1 (antisymmetric eigenspace). The operator $R_i(p) = pT_i + (1-p)$ has eigenvalues:*

$$\lambda_+ = 1 - p(1-q) = \frac{1+q}{2}\Big|_{p=1/2} \qquad \textit{on the } q\textit{-eigenspace of } T_i, \tag{8}$$

$$\lambda_- = 1 - 2p = 0\Big|_{p=1/2} \qquad \textit{on the } (-1)\textit{-eigenspace of } T_i. \tag{9}$$

At $p=1/2$, $R_i(1/2)$ annihilates the antisymmetric eigenspace for all *values of q, and scales the symmetric eigenspace by $(1+q)/2$.*

Proof. If $T_i v = qv$, then $R_i(p)v = (pq + 1 - p)v = (1 - p(1 - q))v$. If $T_i v = -v$, then $R_i(p)v = (-p + 1 - p)v = (1 - 2p)v$. Setting $p = 1/2$ gives the stated values. □

Remark 11. The cactus midpoint $p = 1/2$ is the unique value of p that kills the antisymmetric eigenspace of *every* R_i, independent of q. However, the product $\prod R_{i_j}(1/2)$ fails to be word-independent at generic q due to the braid obstruction (Theorem **??**). Each individual $R_i(1/2)$ is a scaled projector, but their composition does not give a well-defined element of the Hecke algebra when $q \neq 0$.

5 The Spectral Parameter

Proposition 12 (Tikhonov's Spectral Parameter)**.** *Define* $\kappa = \kappa(p, q) := \frac{1-qp}{1-p}$. *Then:*

(a) The random sorting permuton $\mu_{p,q}$ *depends on* (p, q) *only through* κ *(Tikhonov, Theorem 1.4 of [**?**]).*

(b) The three distinguished loci are:

Level	p	κ	*Operator*
Identity	0	1	id
Cactus	1/2	$2 - q$	$(1/2)^m \pi_{w_0}$
Hecke	1	$\frac{1}{1-q}$	T_{w_0}

(c) At $q = 0$*, the cactus level is* $\kappa = 2$*, independent of all other parameters. This is a universal constant: the midpoint between* $\kappa = 1$ *(identity) and* $\kappa = \infty$ *(fully sorted), measured on the scale* $\kappa/(1 + \kappa)$ *which maps* $[1, \infty] \to [1/2, 1]$.

Proof. Part (a) is Tikhonov's theorem. For (b), direct computation: at $p = 0$: $\kappa = 1/1 = 1$; at $p = 1/2$: $\kappa = (1 - q/2)/(1/2) = 2 - q$; at $p = 1$: $\kappa = (1 - q)/0$, interpreted as the limit $\lim_{p\to 1} \kappa = +\infty$ when $q \neq 1$, or via $1/(1 - q)$ in the algebraic sense.

For (c), at $q = 0$, $\kappa(1/2, 0) = 2$. □

6 Why $q = 0$ is Special

We collect the algebraic reasons why the cactus midpoint theorem is special to $q = 0$.

Theorem 13. *The following properties hold at* $q = 0$ *and fail at generic* q*:*

1. ***Idempotency****:* $\pi_i^2 = \pi_i$ *(Lemma* **??***).*
2. ***Braid relation****:* $\pi_i \pi_{i+1} \pi_i = \pi_{i+1} \pi_i \pi_{i+1}$ *(Theorem* **??***).*
3. ***Word-independence****:* π_w *is well-defined (Corollary* **??***).*
4. ***Rank collapse****: The image of* π_{w_0} *acting on the regular representation has dimension* 1 *(all of* $H_0(S_k)$ *maps onto the span of* $\sum_{v \leq w_0} T_v$*).*

The chain of implications is: $(1) + (2) \Rightarrow (3) \Rightarrow (4)$.

Proof. Properties (1) and (2) are proved in Lemma **??** and Theorem **??**. Property (3) follows from (1), (2), and Matsumoto's theorem as in Corollary **??**.

For (4): At $q = 0$, π_i is an idempotent, hence its image is a proper subspace. In the regular representation on basis $\{T_w : w \in S_k\}$, the operator $\pi_i = T_i + 1$ maps T_w to $T_{s_i w} + T_w$ if $\ell(s_i w) > \ell(w)$, and to 0 if $\ell(s_i w) < \ell(w)$ (since $T_i T_w = -T_w$ in H_0 when $\ell(s_i w) < \ell(w)$). The product π_{w_0} maps every basis element to $\sum_{v \in S_k} T_v$, giving rank 1.

To see this, note that π_{w_0} is the unique maximal element in the 0-Hecke monoid: applying all sorting operators in sequence sorts any permutation to the identity, and the image of any T_w under π_{w_0} is $\sum_v T_v$ (the "fully sorted" element).

At generic $q \neq 0$, property (1) fails ($\pi_i^2 = (q+1)\pi_i \neq \pi_i$), (2) fails (obstruction $q(T_i - T_{i+1}) \neq 0$), and in $H_q(S_3)$, the product $(T_1+1)(T_2+1)(T_1+1)$ has rank 3 in the regular representation (verified computationally at $q = 1$). □

7 Computational Verification

All results have been verified computationally:

1. $H_q(S_3)$ **regular representation** (6-dimensional): Hecke relations, quasi-idempotency, braid obstruction $q(T_i - T_{i+1})$, word-dependence at generic q, word-independence at $q = 0$.

2. $H_0(S_3)$ **and** $H_0(S_4)$: Cactus midpoint identity $\prod R_{i_j}(1/2) = (1/2)^m \pi_{w_0}$ verified for all parabolic sub-products.

3. **Rank analysis**: $\mathrm{rank}(\pi_1\pi_2\pi_1) = 1$ at $q = 0$, $= 3$ at $q = 1$, in $H_q(S_3)$.

4. **Spectral parameter**: $\kappa(1/2, q) = 2 - q$ confirmed symbolically.

Scripts: `braid_obstruction_verify.py`, `tikhonov_cactus_test.py`, `cactus_midpoint_generic_q.py`.

References

[1] K. Tikhonov, *Random sorting networks and the Hecke algebra*, arXiv:2604.06532, 2026.